Namal University, Mianwali

DEPARTMENT OF COMPUTER SCIENCES

Virtual Fitness Trainer

Database Design Document

Version 1.0

Submitted By:

Muhammad Wasif	NUM-BSCS-2024-59
Hifza Bibi	NUM-BSCS-2024-27

Submission Date:

15th April, 2026

Revision History

Date	Version	Description of Changes	Approved by
15 April, 2026	1.0	Initial version - Complete Database Design Document	

Contents

1 Project Overview

1.1 Introduction

Virtual Fitness Trainer is an intelligent web-based fitness guidance system that provides personalized workout and diet plans using Artificial Intelligence. The system is designed to help individuals achieve their fitness goals by considering their personal health data, fitness level, preferences, and any existing medical conditions.

This application targets fitness enthusiasts, beginners, and health-conscious individuals who want customized guidance without the need for expensive personal trainers or generic one-size-fits-all programs. The system supports comprehensive features including AI-powered plan generation, exercise video demonstrations, daily progress tracking, and adaptive recommendations.

1.2 Problem Statement

In today's fast-paced world, people struggle to maintain consistent fitness routines due to lack of personalized guidance, time constraints, and limited access to professional trainers. Existing fitness applications often provide generic workout and diet plans that do not consider individual health conditions, body metrics, or specific goals, which can lead to ineffective results or even injuries.

Moreover, manual tracking of progress, lack of proper form guidance through videos, and absence of adaptive plan modification based on user performance are major challenges. There is a clear need for an intelligent, accessible, and personalized fitness solution that can safely guide users according to their unique requirements while continuously adapting to their progress.

1.3 Project Objectives

General Objective:
To develop a comprehensive database-backed Virtual Fitness Trainer system that delivers personalized, AI-powered workout and diet plans with effective progress tracking and adaptive recommendations.

Specific Objectives:

- Enable users to create detailed profiles including health conditions, body measurements, and fitness goals for accurate personalization.
- Generate customized workout and diet plans using AI based on user data and preferences.
- Provide video demonstrations for exercises to ensure correct form and reduce injury risk.
- Allow users to log daily progress and generate insightful weekly and monthly reports.
- Implement adaptive recommendation system that modifies plans based on user performance and feedback.
- Maintain complete audit logs and progress history for better user accountability and system improvement.
- Ensure data integrity and efficient retrieval for seamless user experience.

1.4 GitHub Repository Link

https://github.com/wasif-cmd/Virtual-Fitness-Trainer

2 Detailed Database Design

2.1 Entity

The following table identifies and defines the core entities central to the Virtual Fitness Trainer database system.

Sr. No	Entity Name	Description
1	User	Central entity representing registered users of the system, storing personal and fitness-related information.
2	Login Activity	Records user login sessions, device information, and security-related login data.
3	Profile Audit Log	Maintains history of changes made to user profiles for security and auditing purposes.
4	Workout Plan	Represents personalized workout plans generated for users based on their goals and fitness level.
5	Workout Day	Defines individual days within a workout plan, including focus area and session details.
6	Workout Plan Exercise	Associative entity linking exercises to specific days in a workout plan with sets, reps, and order.
7	Exercise	Stores information about individual exercises including name, difficulty, tips, and target muscle groups.
8	Exercise Video	Contains video resources and demonstration links for exercises.
9	Diet Plan	Represents personalized diet plans generated according to user's caloric needs and preferences.
10	Diet Day	Defines daily meal structure within a diet plan.
11	Meal	Individual meals within a diet day containing nutritional information.
12	Recipe	Detailed cooking instructions and preparation information for meals.
13	Ingredient	Stores nutritional information of food ingredients used in recipes.
14	Meal Ingredient	Associative entity linking ingredients to meals with quantity and unit.
15	Recommendation	AI-generated recommendations for plan adjustments or alternatives.
16	Progress Log	Records user's daily progress including weight, measurements, and workout completion.
17	Progress Log Exercise	Tracks performance details (sets, reps, weight) for individual exercises in progress logs.

2.2 Data Dictionary

This section provides detailed attribute definitions for each entity.

2.2.1 User

Sr. No	Attribute Name	Data Type	Constraint	Description
1	User ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each user.
2	Name	VARCHAR(100)	NOT NULL	Full name of the user.
3	Email	VARCHAR(100)	NOT NULL, UNIQUE	User's email address used for login and notifications.
4	Password Hash	VARCHAR(255)	NOT NULL	Securely hashed password for authentication.
5	Birth Date	DATE	NOT NULL	User's date of birth.
6	Gender	ENUM	NOT NULL	User's gender.
7	Height	DECIMAL(5,2)	NOT NULL	Height in centimeters.
8	Weight	DECIMAL(5,2)	NOT NULL	Current weight in kilograms.
9	Body Type	VARCHAR(50)	NULL	User's body type (e.g., Ectomorph, Mesomorph, Endomorph).
10	Primary Goal	VARCHAR(100)	NOT NULL	Main fitness goal (e.g., Weight Loss, Muscle Gain, Maintenance).
11	Target Weight	DECIMAL(5,2)	NULL	Desired target weight in kilograms.
12	Timeline	VARCHAR(50)	NULL	Desired timeline to achieve goal.
13	Current Fitness Level	ENUM	NOT NULL	User's current fitness level.
14	Workout Frequency	INT	NOT NULL	Number of workout days per week.
15	Age	INT	NOT NULL	Calculated or provided age of the user.
16	BMI	DECIMAL(4,2)	NULL	Body Mass Index (calculated).
17	BMR	DECIMAL(6,2)	NULL	Basal Metabolic Rate (calculated).
18	TDEE	DECIMAL(6,2)	NULL	Total Daily Energy Expenditure (calculated).
19	Health Conditions	TEXT	NULL	Any existing medical conditions or injuries.
20	Injuries	TEXT	NULL	Specific injuries or physical limitations.
21	Dietary Preference	VARCHAR(100)	NULL	Dietary preferences (e.g., Vegetarian, Vegan, Keto).
22	Food Allergy	TEXT	NULL	List of food allergies.

23	Created At	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Account creation timestamp.
24	Updated At	TIMESTAMP	NULL	Last profile update timestamp.

2.2.2 Login Activity

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Login ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each login record.
2	Email	VARCHAR(100)	NOT NULL	Email address used during login.
3	Remember Me	BOOLEAN	NULL	Indicates if "Remember Me" option was selected.
4	Is Failed	BOOLEAN	NOT NULL, DEFAULT FALSE	Flag to indicate whether login attempt failed.
5	Login Time	TIMESTAMP	NOT NULL	Timestamp of the login attempt.
6	IP Address	VARCHAR(45)	NOT NULL	IP address of the user during login.
7	Device Category	VARCHAR(50)	NULL	Type of device used (Mobile, Desktop, Tablet).
8	OS Name	VARCHAR(50)	NULL	Operating system name (e.g., Android, iOS, Windows).
9	OS Version	VARCHAR(50)	NULL	Operating system version.
10	Browser Name	VARCHAR(50)	NULL	Browser name used for login.
11	Browser Version	VARCHAR(50)	NULL	Browser version.
12	User ID	UUID	FOREIGN KEY, NOT NULL	References the User entity.

2.2.3 Profile Audit Log

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Audit ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each profile audit record.
2	User ID	UUID	FOREIGN KEY, NOT NULL	References the User whose profile was changed.
3	Timestamp	TIMESTAMP	NOT NULL	Date and time when the change occurred.

4	IP Address	VARCHAR(45)	NOT NULL	IP address from which the change was made.
5	Field Changed	VARCHAR(100)	NOT NULL	Name of the field that was modified.
6	Old Value	TEXT	NULL	Previous value of the field before change.
7	New Value	TEXT	NULL	New value of the field after change.

2.2.4 Workout Plan

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Workout Plan ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each workout plan.
2	User ID	UUID	FOREIGN KEY, NOT NULL	References the User who owns this plan.
3	Title	VARCHAR(150)	NOT NULL	Title or name of the workout plan.
4	Goal	VARCHAR(100)	NOT NULL	Primary fitness goal of the plan (e.g., Weight Loss, Muscle Gain).
5	Created At	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Date and time when the plan was created.
6	Activation Status	BOOLEAN	NOT NULL, DEFAULT TRUE	Whether the plan is currently active.
7	Plan Type	VARCHAR(50)	NOT NULL	Type of plan (e.g., Beginner, Intermediate, Advanced).
8	Start Date	DATE	NOT NULL	Start date of the workout plan.
9	End Date	DATE	NULL	Expected end date of the plan.

2.2.5 Workout Day

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Workout Day ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each day in the workout plan.
2	Workout Plan ID	UUID	FOREIGN KEY, NOT NULL	References the parent Workout Plan.
3	Day Number	INT	NOT NULL	Sequential day number in the plan (1, 2, 3...).

4	Day Name	VARCHAR(50)	NULL	Name of the day (e.g., Push Day, Leg Day).
5	Focus Area	VARCHAR(100)	NOT NULL	Primary muscle group or focus area for the day.
6	Session Duration	INT	NULL	Estimated session duration in minutes.
7	Calories Estimate	DECIMAL(6,2)	NULL	Estimated calories to be burned during the session.
8	Notes	TEXT	NULL	Additional notes or instructions for the day.
9	Is Rest Day	BOOLEAN	NOT NULL, DEFAULT FALSE	Indicates if this is a rest/recovery day.

2.2.6 Workout Plan Exercise

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Plan Exercise ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for the exercise assignment.
2	Workout Day ID	UUID	FOREIGN KEY, NOT NULL	References the Workout Day.
3	Exercise ID	UUID	FOREIGN KEY, NOT NULL	References the Exercise entity.
4	Sets	INT	NOT NULL	Number of sets to perform.
5	Reps	VARCHAR(50)	NOT NULL	Number of repetitions (e.g., 8-12, 10).
6	Rest Time Seconds	INT	NULL	Rest time between sets in seconds.
7	Intensity	VARCHAR(50)	NULL	Intensity level (e.g., Moderate, High).
8	Notes	TEXT	NULL	Specific instructions or notes for the exercise.
9	Order In Day	INT	NOT NULL	Order/sequence of the exercise in the day.

2.2.7 Exercise

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Exercise ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each exercise.
2	Exercise Name	VARCHAR(150)	NOT NULL	Name of the exercise.
3	Description	TEXT	NOT NULL	Detailed description of how to perform the exercise.

4	Difficulty Level	ENUM	NOT NULL	Difficulty level of the exercise.
5	Tips	TEXT	NULL	Helpful tips for proper execution.
6	Equipment Needed	VARCHAR(200)	NULL	Equipment required for the exercise.
7	Exercise Category	VARCHAR(100)	NULL	Category of exercise (e.g., Strength, Cardio, Flexibility).
8	Muscle Group	VARCHAR(100)	NOT NULL	Primary muscle group targeted.
9	Target Area	VARCHAR(100)	NULL	Specific target area (e.g., Upper Chest, Quads).
10	Common Mistakes	TEXT	NULL	Common mistakes to avoid.
11	Calories Burned Per Rep	DECIMAL(5,2)	NULL	Estimated calories burned per repetition.

2.2.8 Exercise Video

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Video ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each exercise video.
2	Exercise ID	UUID	FOREIGN KEY, NOT NULL	References the Exercise entity.
3	URL	VARCHAR(255)	NOT NULL	Direct URL to the exercise demonstration video.
4	Embedded Code	TEXT	NULL	HTML embed code for video player.
5	Description	TEXT	NULL	Description or instructions related to the video.
6	Language	VARCHAR(50)	NULL	Language of the video (e.g., English, Urdu).

2.2.9 Progress Log

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Log ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each progress log entry.
2	User ID	UUID	FOREIGN KEY, NOT NULL	References the User entity.
3	Assigned Workout Day ID	UUID	FOREIGN KEY, NULL	References the Workout Day being logged (if applicable).

4	Log Date	DATE	NOT NULL	Date of the progress log entry.
5	Weight	DECIMAL(5,2)	NULL	Current body weight in kilograms.
6	Chest	DECIMAL(5,2)	NULL	Chest measurement in cm.
7	Hips	DECIMAL(5,2)	NULL	Hips measurement in cm.
8	Waist	DECIMAL(5,2)	NULL	Waist measurement in cm.
9	Arms	DECIMAL(5,2)	NULL	Arms measurement in cm.
10	Thighs	DECIMAL(5,2)	NULL	Thighs measurement in cm.
11	Energy Level	INT	NULL	Self-reported energy level (1-10).
12	Mood	VARCHAR(50)	NULL	Self-reported mood (e.g., Energized, Tired, Motivated).
13	Workouts Completed	INT	NULL	Number of workouts completed.
14	Notes	TEXT	NULL	Additional notes or reflections by the user.
15	Created At	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Timestamp when the log was created.

2.2.10 Progress Log Exercise

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Progress Log Exercise ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for progress on a specific exercise.
2	Log ID	UUID	FOREIGN KEY, NOT NULL	References the Progress Log.
3	Exercise ID	UUID	FOREIGN KEY, NOT NULL	References the Exercise entity.
4	Sets Completed	INT	NULL	Number of sets actually completed.
5	Reps Completed	VARCHAR(50)	NULL	Repetitions completed (e.g., 10, 8-12).
6	Notes	TEXT	NULL	Notes regarding performance or difficulty.

2.2.11 Diet Plan

Sr. No	Attribute Name	Data Type	Constraint	Description

1	Diet Plan ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each diet plan.
2	User ID	UUID	FOREIGN KEY, NOT NULL	References the User who owns this diet plan.
3	Title	VARCHAR(150)	NOT NULL	Title or name of the diet plan.
4	Plan Type	VARCHAR(50)	NOT NULL	Type of diet plan (e.g., Weight Loss, Muscle Gain).
5	Daily Calorie Target	INT	NOT NULL	Target daily calorie intake.
6	Protein Target	DECIMAL(6,2)	NULL	Target daily protein intake in grams.
7	Carb Target	DECIMAL(6,2)	NULL	Target daily carbohydrate intake in grams.
8	Fat Target	DECIMAL(6,2)	NULL	Target daily fat intake in grams.
9	Duration Weeks	INT	NOT NULL	Duration of the plan in weeks.
10	Start Date	DATE	NOT NULL	Start date of the diet plan.
11	End Date	DATE	NULL	Expected end date of the plan.
12	Status	VARCHAR(50)	NOT NULL	Current status of the plan (Active, Completed, Paused).
13	Cultural Preference	VARCHAR(100)	NULL	Cultural or cuisine preference (e.g., Pakistani, Asian).
14	Budget Level	VARCHAR(50)	NULL	Budget level (Low, Medium, High).
15	Created At	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Timestamp when the plan was created.
16	Updated At	TIMESTAMP	NULL	Last update timestamp.

2.2.12 Diet Day

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Diet Day ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each day in the diet plan.
2	Diet Plan ID	UUID	FOREIGN KEY, NOT NULL	References the parent Diet Plan.
3	Day Number	INT	NOT NULL	Sequential day number in the plan.
4	Day Name	VARCHAR(50)	NULL	Name of the day (e.g., Day 1, Monday).

5	Total Calories	INT	NOT NULL	Total calories allocated for the day.
6	Notes	TEXT	NULL	Additional notes or instructions for the day.

2.2.13 Meal

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Meal ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each meal.
2	Diet Day ID	UUID	FOREIGN KEY, NOT NULL	References the Diet Day.
3	Meal Type	VARCHAR(50)	NOT NULL	Type of meal (Breakfast, Lunch, Dinner, Snack).
4	Meal Name	VARCHAR(150)	NOT NULL	Name of the meal.
5	Calories	INT	NOT NULL	Calories in the meal.
6	Protein	DECIMAL(5,2)	NULL	Protein content in grams.
7	Carbs	DECIMAL(5,2)	NULL	Carbohydrate content in grams.
8	Fats	DECIMAL(5,2)	NULL	Fat content in grams.
9	Order In Day	INT	NOT NULL	Sequence order of the meal in the day.
10	Notes	TEXT	NULL	Special instructions or notes for the meal.

2.2.14 Recipe

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Recipe ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each recipe.
2	Meal ID	UUID	FOREIGN KEY, NOT NULL	References the Meal.
3	Instructions	TEXT	NOT NULL	Step-by-step cooking instructions.
4	Preparation Time	INT	NULL	Preparation time in minutes.
5	Cooking Time	INT	NULL	Cooking time in minutes.

2.2.15 Ingredient

Sr. No	Attribute Name	Data Type	Constraint	Description

1	Ingredient ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each ingredient.
2	Ingredient Name	VARCHAR(100)	NOT NULL	Name of the ingredient.
3	Unit	VARCHAR(50)	NOT NULL	Measurement unit (e.g., gram, ml, piece).
4	Calories Per Unit	DECIMAL(6,2)	NOT NULL	Calories per unit of the ingredient.

2.2.16 Meal Ingredient

Sr. No	Attribute Name	Data Type	Constraint	Description
1	MI ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for meal-ingredient relationship.
2	Meal ID	UUID	FOREIGN KEY, NOT NULL	References the Meal.
3	Ingredient ID	UUID	FOREIGN KEY, NOT NULL	References the Ingredient.
4	Quantity	DECIMAL(8,2)	NOT NULL	Quantity of the ingredient used.

2.2.17 Recommendation

Sr. No	Attribute Name	Data Type	Constraint	Description
1	Recommendation ID	UUID	PRIMARY KEY, NOT NULL	Unique identifier for each recommendation.
2	User ID	UUID	FOREIGN KEY, NOT NULL	References the User for whom the recommendation is generated.
3	Recommendation Type	VARCHAR(50)	NOT NULL	Type of recommendation (Workout, Diet, General).
4	Trigger Reason	TEXT	NOT NULL	Reason why this recommendation was triggered.
5	Recommendation Date	DATE	NOT NULL	Date when the recommendation was generated.
6	Status	VARCHAR(50)	NOT NULL	Status of recommendation (Pending, Accepted, Rejected).
7	User Response Date	DATE	NULL	Date when user responded to the recommendation.
8	User Feedback	TEXT	NULL	Feedback provided by the user.
9	Suggested Changes	TEXT	NULL	Suggested modifications to plans.

10	Source Diet Plan ID	UUID	FOREIGN KEY, NULL	References source Diet Plan (if applicable).
11	Source Workout Plan ID	UUID	FOREIGN KEY, NULL	References source Workout Plan (if applicable).

2.3 Business Rules

The following business rules define the operational constraints, data integrity requirements, and system behavior for the Virtual Fitness Trainer application:

1. Each `User` must have a unique email address and a securely hashed password for authentication.
2. A user can have only one active `Workout Plan` and one active `Diet Plan` at any given time.
3. A `Workout Plan` must contain at least one `Workout Day`, and each `Workout Day` may contain exercises through the `Workout Plan Exercise` associative entity.
4. The system shall automatically calculate and store `BMI`, `BMR`, and `TDEE` whenever a user updates their height, weight, age, or gender.
5. Workout plans and diet plans must be generated considering the user's `Health Conditions`, `Injuries`, `Dietary Preference`, and `Food Allergy`.
6. A `Progress Log` can only be created for the current date or past dates. Future dates are not allowed.
7. Every exercise in a workout plan can have multiple associated `Exercise Videos` for proper demonstration but one `Exercise Video` must belong to exaclty one `Exercise`.
8. When a user completes a workout, the system must record performance details through the `Progress Log Exercise` entity (sets completed, reps, weight used, etc.).
9. The `Recommendation` engine shall generate suggestions when a user's progress deviates significantly from the plan targets or when health metrics change.
10. All profile modifications must be logged in the `Profile Audit Log` with the user ID, timestamp, IP address, and before/after values for security and accountability.
11. Every login attempt (successful or failed) must be recorded in the `Login Activity` table with timestamp and device information.
12. A `Diet Plan` must maintain nutritional balance by ensuring total calories and macros (Protein, Carbs, Fat) are properly distributed across meals in `Diet Day`.
13. Users cannot delete their account if they have active workout or diet plans. They must first archive or complete the plans.
14. The system must enforce referential integrity between all entities. For example, a `Workout Plan Exercise` cannot exist without a valid `Exercise` and `Workout Day`.
15. Progress reports (weekly/monthly) shall be generated based on data from `Progress Log` and `Progress Log Exercise` entities.

2.4 Entity Relationship Diagram (Crow's Foot Notation)

The following ERD represents the complete logical database design for the Virtual Fitness Trainer system using Crow's Foot notation. It illustrates all major entities, their attributes, primary keys, foreign keys, and the cardinalities of relationships between them.

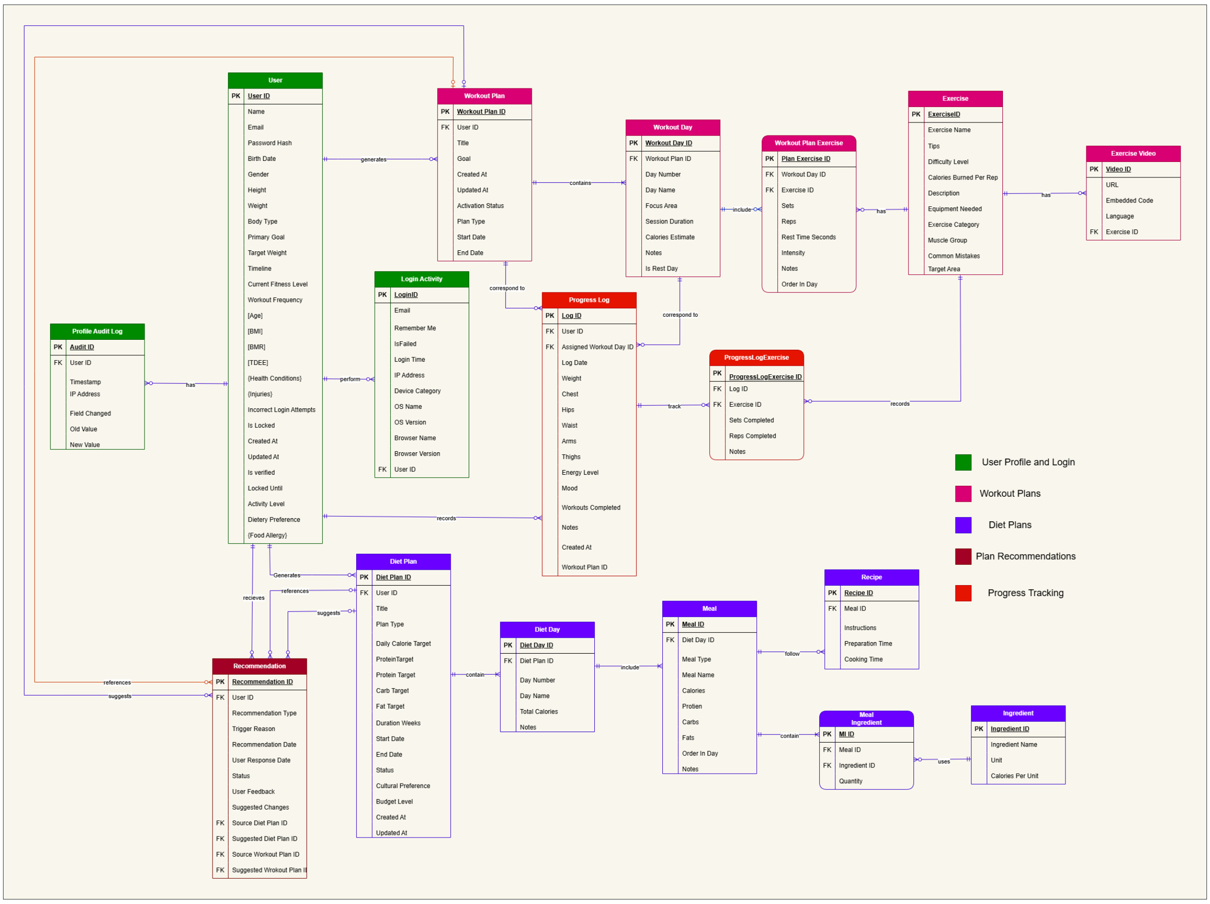

Figure 1: Virtual Fitness Trainer Entity Relationship Diagram (Crow's Foot Notation)

ERD Description:

The ERD consists of the following key relationships and cardinalities:

- **User** is the central entity and has a one-to-many relationship with `Login Activity`, `Profile Audit Log`, `Workout Plan`, `Diet Plan`, `Progress Log`, and `Recommendatio`
- One `Workout Plan` **contains** many `Workout Days` (1:N).
- One `Workout Day` **includes** many `Workout Plan Exercises` (1:N).
- `Workout Plan Exercise` is an associative entity resolving the many-to-many relationship between `Workout Day` and `Exercise` with additional attributes (sets, reps, order, etc.).
- One `Exercise` **has** one or more `Exercise Videos` (1:N).
- One `Diet Plan` **contains** many `Diet Days` (1:N).
- One `Diet Day` **includes** many `Meals` (1:N).
- One `Meal` **uses** many `Ingredients` through the associative entity `Meal Ingredient` (M:N).
- One `Meal` **follows** one `Recipe` (1:1).

- One `Progress Log` **tracks** many `Progress Log Exercise` entries (1:N).
- One `User` **receives** many `Recommendations` generated by the AI system (1:N).
- `Recommendation` may reference source `Workout Plan` and/or `Diet Plan`.

Additional Notation:

- **PK** = Primary Key
- **FK** = Foreign Key
- Green boxes = User Profile & Login related entities
- Pink boxes = Workout Plans related entities
- Blue boxes = Diet Plans related entities
- Light Red boxes = Progress Tracking
- Dark Red boxes = Recommendations

3 References

1. M. D. Mifflin, S. T. St Jeor, L. A. Hill, B. J. Scott, S. A. Daugherty, and Y. O. Koh, "A new predictive equation for resting energy expenditure in healthy individuals," *American Journal of Clinical Nutrition*, vol. 51, no. 2, pp. 241–247, Feb. 1990.

2. Muhammad Wasif, Muhammad Shahid, and Hifza Bibi, "Software Requirements Specification for Virtual Fitness Trainer," Version 2.0, Department of Computer Science, Namal University, Mianwali, Pakistan, Dec. 2025.

3. Project Proposal: Virtual Fitness Trainer, Department of Computer Science, Namal University, Mianwali, Nov. 2025.

4. BodBot, "BodBot Personal Trainer," Google Play Store, 2025. [Online]. Available: https://play.google.com/store/apps/details?id=com.bodbot

5. Leap Fitness Group, "Home Workout – No Equipment," Google Play Store, 2025.

6. Anthropic, "Claude," Large language model, 2025. [Online]. Available: https://www.anthropic.com

7. xAI, "Grok," Large language model, 2025. [Online]. Available: https://x.ai

4 Appendix: AI Usage and Prompts

This section documents the use of AI tools during the development of the Database Design Document for the Virtual Fitness Trainer project.

AI Tools Used:

- Grok (by xAI)
- Claude (by Anthropic)

Purpose of AI Usage: AI was used to assist in database modeling, ERD refinement, business rules formulation, data dictionary creation, and generating professional LaTeX documentation according to the provided template.

Sr. No	Prompt	Purpose
1	You are an expert sql database engineer... i have provided you some documents: 1.SRS of virtual fitness trainer.. 2.VFT-ERD.drawio... you have to assist me in designing ERD according to my project and real world scenarios...	Initial discussion and guidance for ERD design.
2	I have a suggestions for this entity: 1.add FK of video id .. 2.addition of attribute common mistakes... 3.does this system needs an admin dashboard...	Refining Exercise and Exercise Video entities + design decisions.
3	...how to model workout plan for 7 days of weeks?	Designing recurring weekly workout plan structure.
4	now check the modeling of workout plan ......is it correct or need some improvements?	Validation and improvement of Workout Plan modeling.
5	now move towards modeling of diet plan for seven days of week having at least 3 meals a day and recipies...	Designing Diet Plan, Diet Day, Meal, and Recipe structure.
6	explain what these attributes indicates: PrepTimeMinutes-CookTimeMinutes-Servings...	Clarification of Recipe entity attributes.
7	cross check the modeling of diet plan... and workout plan in a best way in erd.	Quality review and refinement of both plans.
8	what attributes can be added in the exercise video table? I have a suggestion to add embedded code...	Enhancing Exercise Video entity.
9	now lets move towards progress tracking module (for FR-6 and FR-7)...	Designing Progress Log and Progress Log Exercise.
10	Keep everything in one ProgressLog table (simple)... what can be done to model the progress log...	Progress tracking module refinement.
11	During entry of progress log attributes can system show list of exercises...	Usability and relationship discussion.
12	what will be the relationship between progress log and workout day.	Defining relationships for progress tracking.

13	Plan Change Recommendations: Every four weeks, the system checks the user's progress... Suggestions are saved in a Recommendations table.	Modeling Recommendation entity.
14	SF-9. User Profile Updates... All profile changes are recorded in the AuditLogs table...	Designing Profile Audit Log.
15	SF-5. Workout Video Demonstration... how to log the watched videos...	Discussion on video watching tracking.
16	cross check this erd against milestone 2, SRS and given project proposal... and provide corrections and suggestions.	Final ERD validation against requirements.
17	I have provided you the four attachments: 1.database design document template... your task is to generate the latex code...	Generation of complete Database Design Document in LaTeX.